Indian Institute of Technology (IIT-Bombay)

AUTUMN Semester, 2026
COMPUTER SCIENCE AND ENGINEERING

CS230: Digital Logic Design and Computer Architecture

Tutorial - III

Full Marks: 0

Time allowed: ∞ hours

1. A processor X_1 operating at 2 GHz has a standard 5-stage RISC instruction pipeline having a base CPI (cycles per instruction) of one without any pipeline hazards. For a given program P that has 30% branch instructions, control hazards incur 2 cycles stall for every branch. A new version of the processor X_2 operating at the same clock frequency has an additional branch predictor unit (BPU) that completely eliminates stalls for correctly predicted branches. There is neither any savings nor any additional stalls for wrong predictions. There are no structural hazards and data hazards for X_1 and X_2. If the BPU has a prediction accuracy of 80%, the speed up (rounded off to two decimal places) obtained by X_2 over X_1 in executing P is _____.

2. A byte-addressable computer system uses a 36-bit memory address and a set-associative cache of size 16 KB. Each cache line contains 64 bytes, and the tag field occupies 24 bits of the memory address. The remaining address bits are used for the set index and the block offset. determine the associativity of the cache.

3. A computer system uses a two-level cache hierarchy consisting of an L1 cache and an L2 cache between the CPU and main memory. Every memory request first accesses the L1 cache, which takes 2 ns. The L1 cache has a hit ratio of 90%. If a request misses in L1, it accesses the L2 cache, which takes an additional 8 ns, and 80% of the requests that reach L2 can get hit. If a request misses in both L1 and L2, main memory is accessed, requiring an additional 50 ns. Assume that main memory always contains the requested data whenever it is accessed. Calculate the average memory access time (AMAT) of the system.

4. A processor has a 5-stage pipeline (IF, ID, EX, MEM, WB). Conditional branches are resolved at the end of the Execute (EX) stage. A branch mis-prediction incurs a penalty of 2 stall cycles.

The processor utilizes a 2-bit saturating counter for branch prediction. The states are represented as:

- **00**: Strongly Not Taken
- **01**: Weakly Not Taken
- **10**: Weakly Taken
- **11**: Strongly Taken

Consider a program executing a loop 50 times. Inside the loop, there is a single conditional branch instruction, `B1`. The actual outcome of `B1` follows a strictly repeating pattern of length 3: **Taken (T), Not Taken (NT), Not Taken (NT)**.

Assume the 2-bit predictor for `B1` is initialized to the state **10 (Weakly Taken)** before the loop begins execution.

Calculate the total number of branch mispredictions and the total pipeline stall cycles incurred solely due to branch `B1` during the 50 iterations of the loop.

5. Consider a classic 5-stage MIPS RISC pipeline with the following stages: Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Write-Back (WB).

The processor supports comprehensive data forwarding (bypassing) to minimize stalls. Specifically, the following forwarding paths are available:

(a) **EX/MEM to EX:** Forwarding ALU results to the ALU input.

(b) **MEM/WB to EX:** Forwarding memory read data or older ALU results to the ALU input.

(c) **MEM/WB to MEM:** Forwarding data directly to the Data Memory input for Store instructions.

Additionally, the register file supports internal forwarding: a write operation in the WB stage occurs in the first half of the clock cycle, while a read in the ID stage occurs in the second half. Pipeline stalls are inserted *only* when a data hazard cannot be resolved using these forwarding paths (e.g., a Load-Use hazard).

Analyze the execution of the following sequence of instructions:

```
I1: LW   R1, 0(R2)      ; R1 <- Memory[R2 + 0]
I2: SUB  R4, R1, R3     ; R4 <- R1 - R3
I3: ADD  R5, R4, R1     ; R5 <- R4 + R1
I4: SW   R5, 0(R1)      ; Memory[R1 + 0] <- R5
I5: LW   R6, 0(R5)      ; R6 <- Memory[R5 + 0]
```

(a) Identify all True Data Dependencies (Read-After-Write) present in this instruction sequence.

(b) Determine the pipeline schedule for these 5 instructions. Clearly state where stall cycles (bubbles) must be inserted and explain the forwarding mechanisms used.

(c) Calculate the total number of clock cycles required to completely execute this instruction sequence.